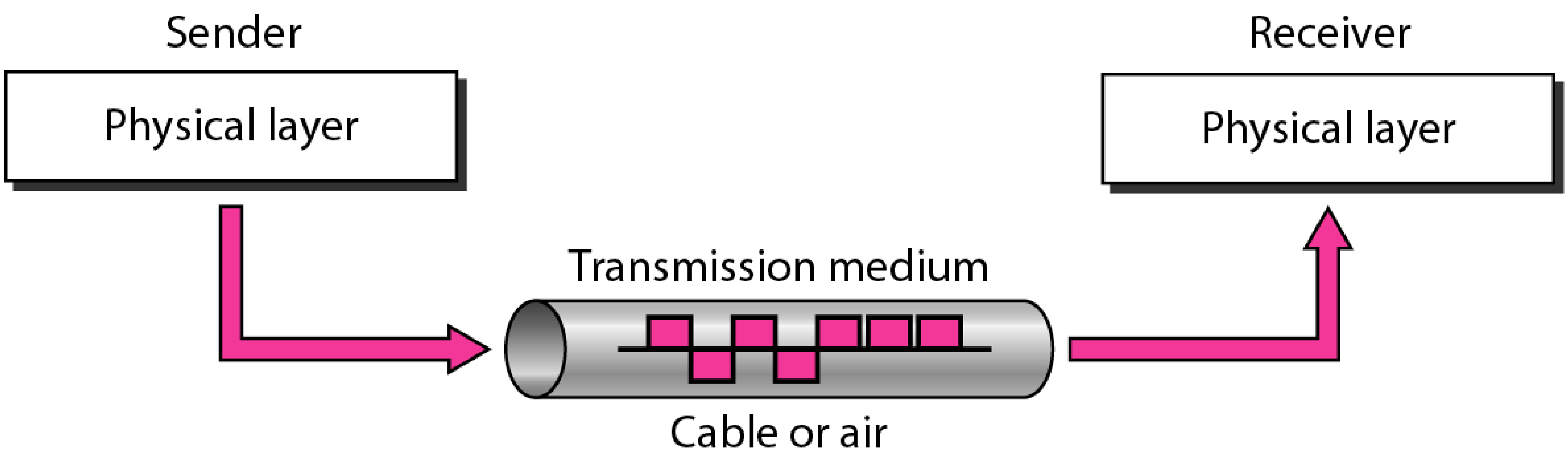
# Transmission Media

# Transmission Media-Introduction

- **Transmission media** are actually located below the physical layer and are directly controlled by the physical layer.
- We could say that transmission media belong to **layer zero**.

# Transmission medium and physical layer

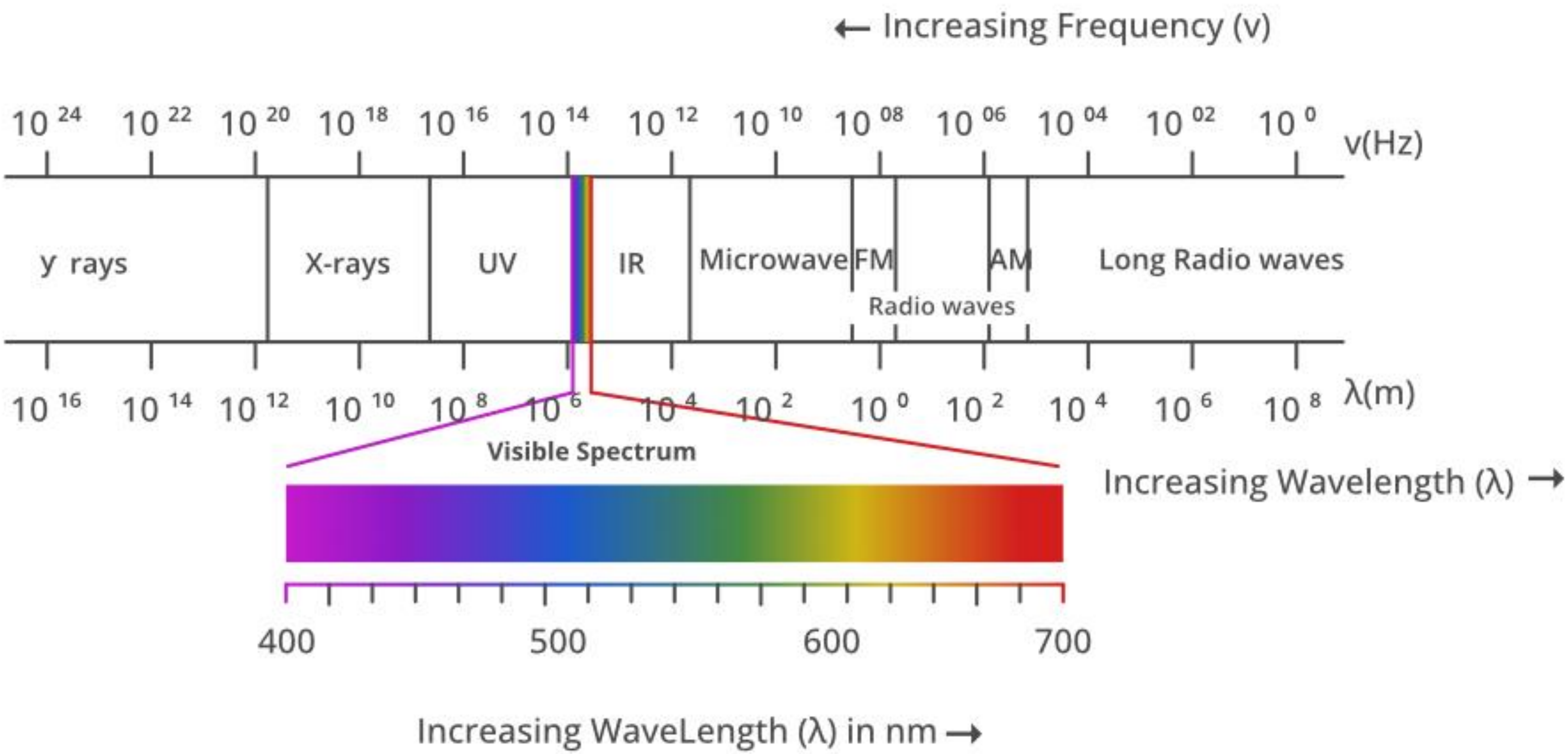


# Transmission Media-Introduction

- A **transmission medium** can be broadly defined as anything that can carry information from a source to a destination.
- In data communications, the *transmission medium is usually free space, metallic cable, or fiber-optic cable*.
- The information is usually a **signal** that is the result of a **conversion of data** from another form.

# Transmission Media-Introduction

- Computers and other telecommunication devices use **signals** to represent data.
- These signals are transmitted from one device to another in the form of **electromagnetic energy**, which is propagated through transmission media.
- **Electromagnetic energy**, a combination of electric and magnetic fields vibrating in relation to each other, includes power, radio waves, infrared light, visible light, ultraviolet light, and X, gamma, and cosmic rays.
- Not all portions of the EM spectrum are currently usable for telecommunications, however.
- The media to harness those that are usable are also limited to a few types.



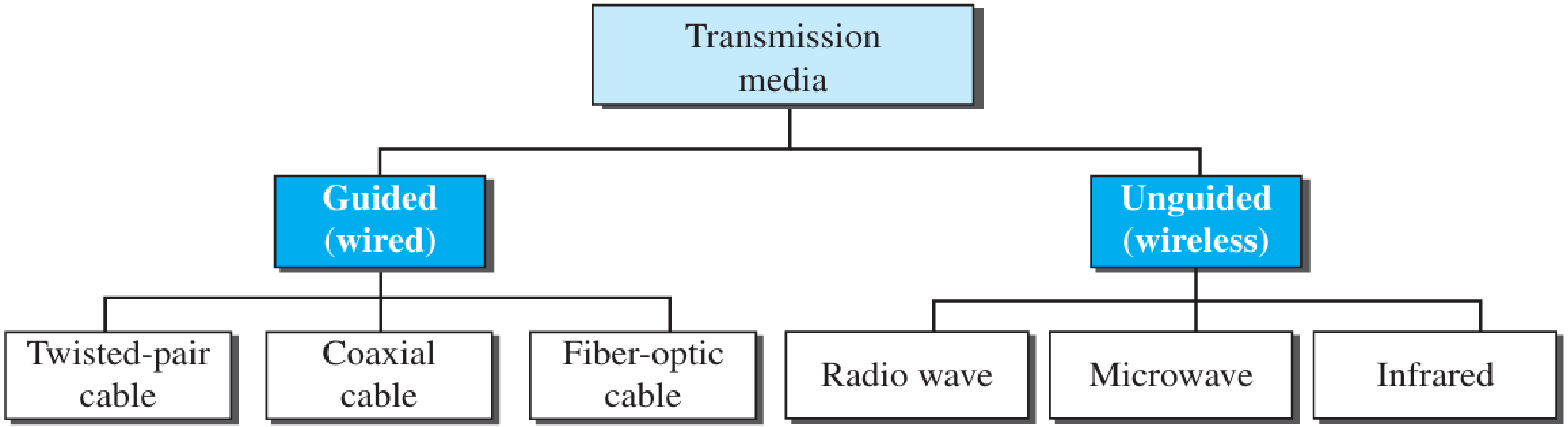
← Increasing Frequency (v)
$10^{24}$ $10^{22}$ $10^{20}$ $10^{18}$ $10^{16}$ $10^{14}$ $10^{12}$ $10^{10}$ $10^{08}$ $10^{06}$ $10^{04}$ $10^{02}$ $10^{0}$
v(Hz)
γ rays
X-rays
UV
IR
Microwave
FM
AM
Long Radio waves
Radio waves
$10^{16}$ $10^{14}$ $10^{12}$ $10^{10}$ $10^{8}$ $10^{6}$ $10^{4}$ $10^{2}$ $10^{0}$ $10^{2}$ $10^{4}$ $10^{6}$ $10^{8}$
λ(m)
Visible Spectrum
Increasing Wavelength (λ) →
400
500
600
700
Increasing WaveLength (λ) in nm →

# Transmission Media-Introduction

- In telecommunications, transmission media can be divided into two broad categories:
  1. guided and
  2. unguided
- Guided media include twisted-pair cable, coaxial cable, and fiber-optic cable.
- Unguided medium is free space.

# Transmission Medium-Introduction

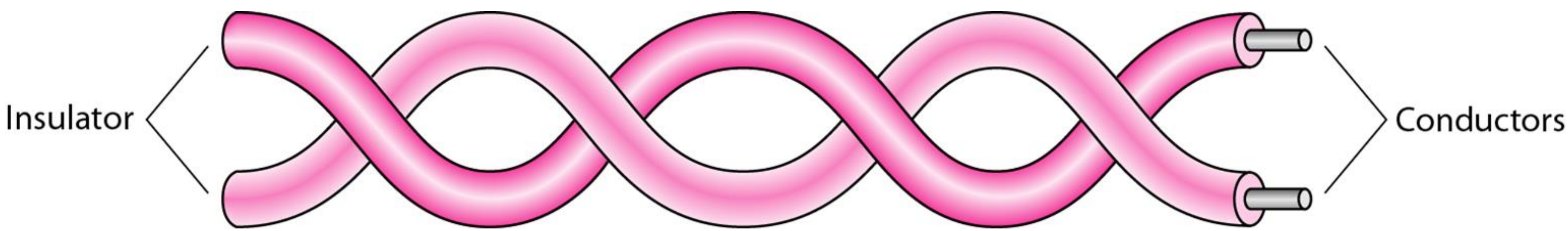
**Figure 7.2** *Classes of transmission media*



# GUIDED MEDIA

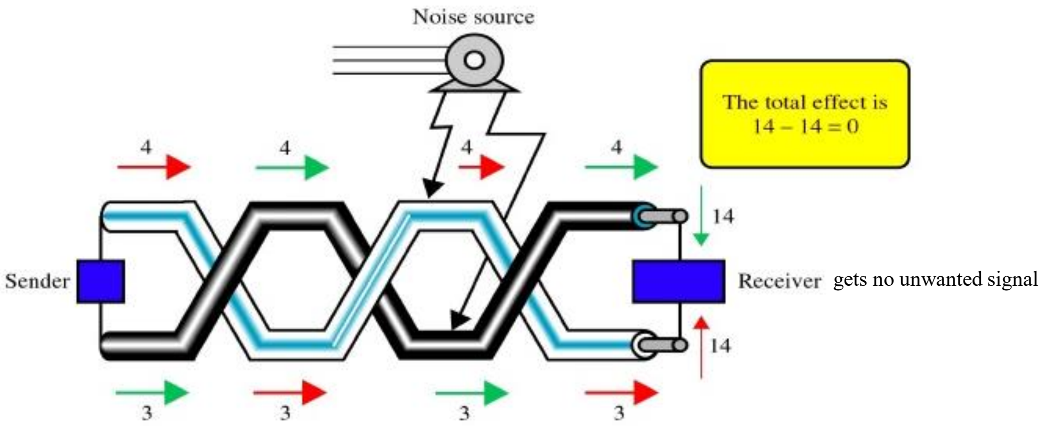
- **Guided media** provide a conduit from one device to another
- It include twisted-pair cable, coaxial cable, and fiber-optic cable.
- A signal traveling along any of these media is directed and contained by the physical limits of the medium.
- **Twisted-pair and coaxial cable** use metallic (copper) conductors that accept and transport signals in the form of electric current
- **Optical fiber** is a cable that accepts and transports signals in the form of light.

# Twisted-pair cable



# Twisted Pair Cable

- Consists of 2 copper conductors(normally copper), each having it's own plastic insulation.
- Two wires are twisted around each other.
- One wire carries electrical signals and the other is ground reference.
- The receiver uses the difference between the two.
- Typically twisted pair is installed in building telephone wiring.
- Twisting is done to reduce the effect of noise on the pair. Each wire is closer to the noise source for half time and farther for the other half.

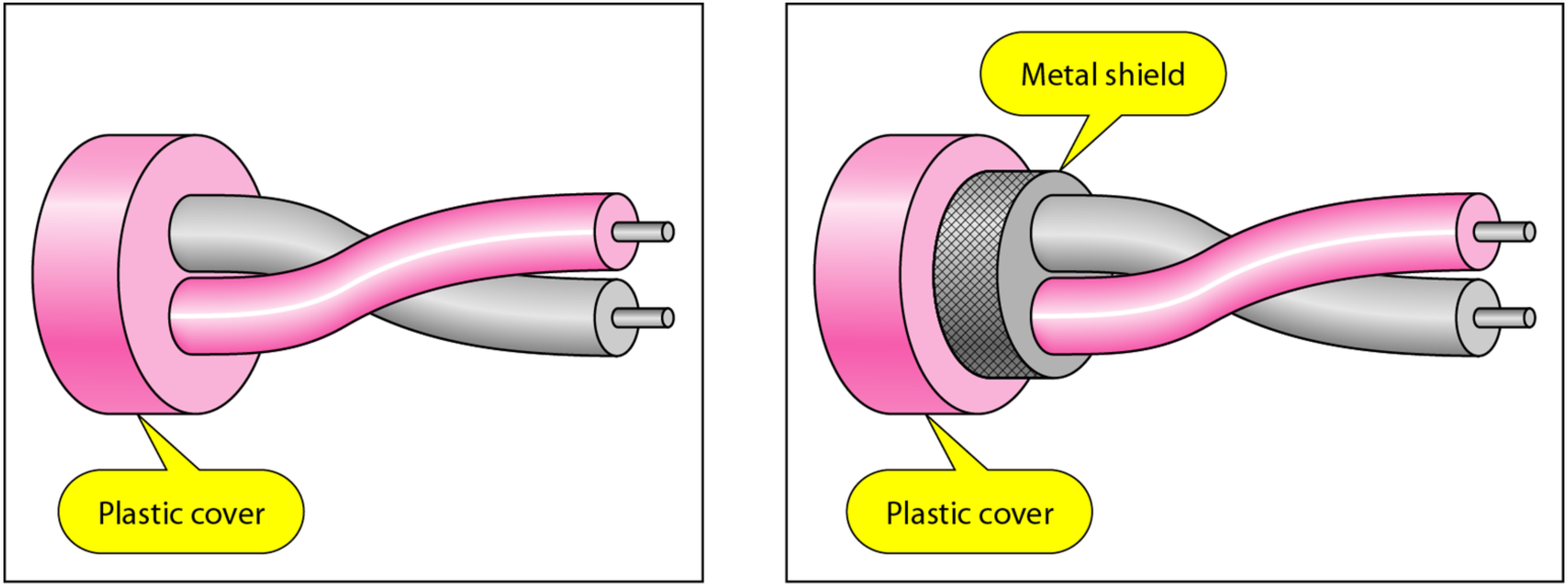


# Twisted Pair Cable

- **Types:**

1. Unshielded twisted pair
2. Shielded twisted pair

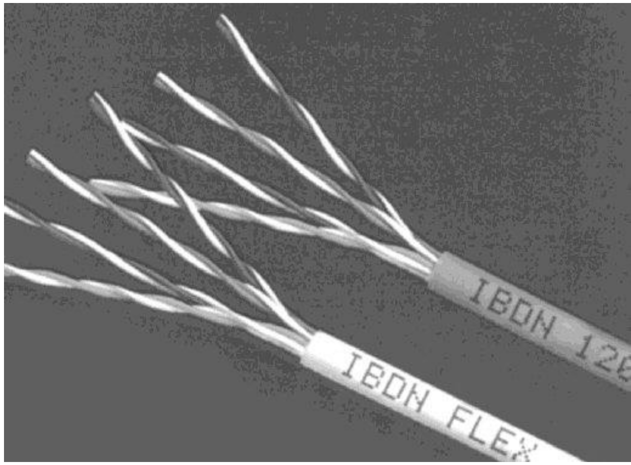
# UTP(Unshielded twisted pair) and STP(Shielded twisted pair) cables



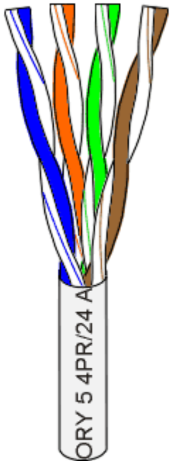

a. UTP    b. STP

# Unshielded Twisted Pair (UTP)

## UTP




- Consists of 4 pairs (8 wires) of insulated copper wires typically about 1 mm thick.
- The wires are twisted together in a helical form.
- Flexible and cheap cable.
- CAT 3, CAT 4, CAT 5, Enhanced CAT 5, CAT 6 and now CAT 7.
- UTP comes in several categories that are based on the cable quality.



# Unshielded Twisted Pair (UTP)

- **Categories**

**Table 7.1** *Categories of unshielded twisted-pair cables*

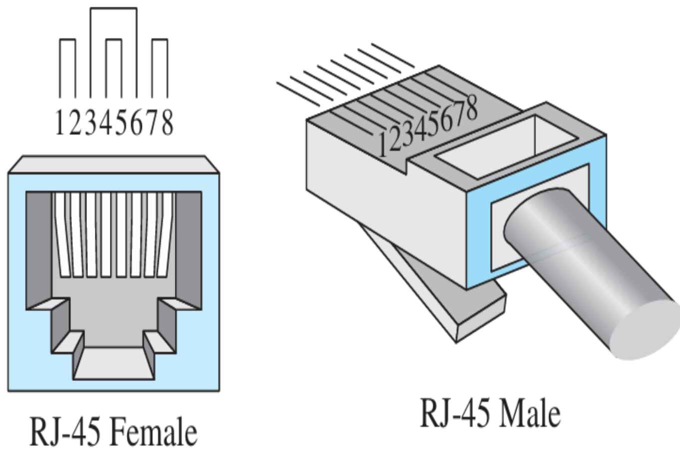
| *Category* | *Specification* | *Data Rate (Mbps)* | *Use* |
|---|---|---|---|
| 1 | Unshielded twisted-pair used in telephone | $< 0.1$ | Telephone |
| 2 | Unshielded twisted-pair originally used in T lines | 2 | T-1 lines |
| 3 | Improved CAT 2 used in LANs | 10 | LANs |
| 4 | Improved CAT 3 used in Token Ring networks | 20 | LANs |
| 5 | Cable wire is normally 24 AWG with a jacket and outside sheath | 100 | LANs |

# Unshielded Twisted Pair (UTP)

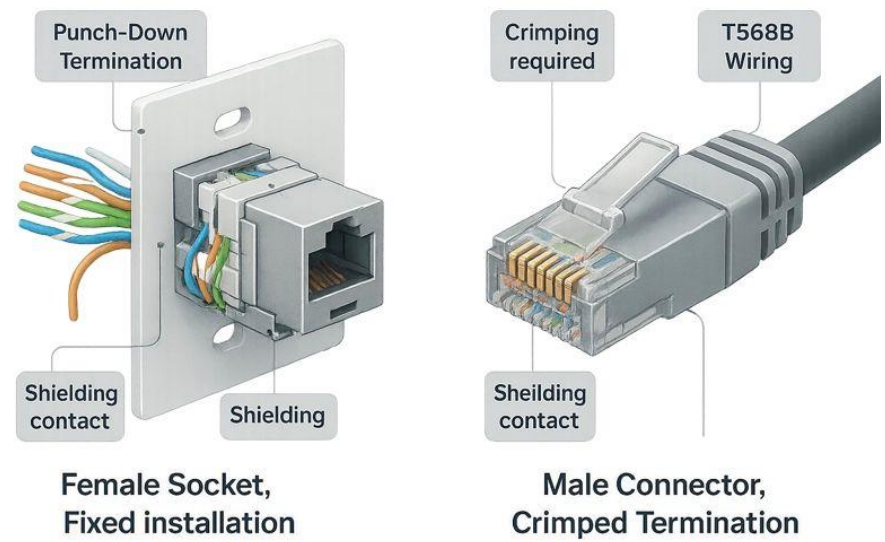
| *Category* | *Specification* | *Data Rate (Mbps)* | *Use* |
|---|---|---|---|
| 5E | An extension to category 5 that includes extra features to minimize the crosstalk and electromagnetic interference | 125 | LANs |
| 6 | A new category with matched components coming from the same manufacturer. The cable must be tested at a 200-Mbps data rate. | 200 | LANs |
| 7 | Sometimes called *SSTP (shielded screen twisted-pair)*. Each pair is individually wrapped in a helical metallic foil followed by a metallic foil shield in addition to the outside sheath. The shield decreases the effect of crosstalk and increases the data rate. | 600 | LANs |

# Unshielded Twisted Pair (UTP)

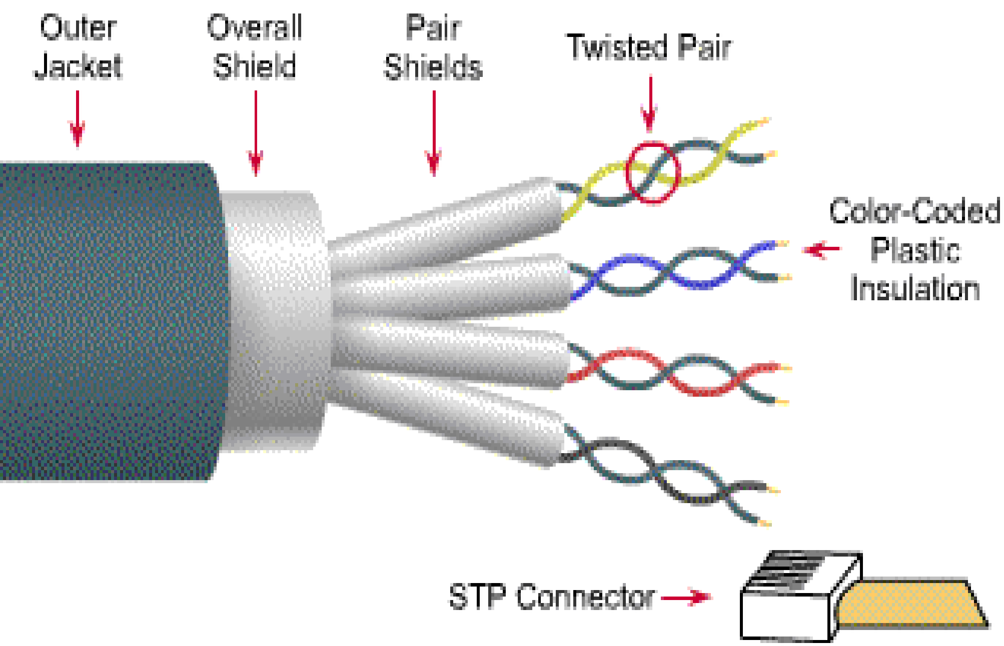
**Figure 7.5** *UTP connector*




- **Connectors**



# Shielded Twisted Pair (STP)

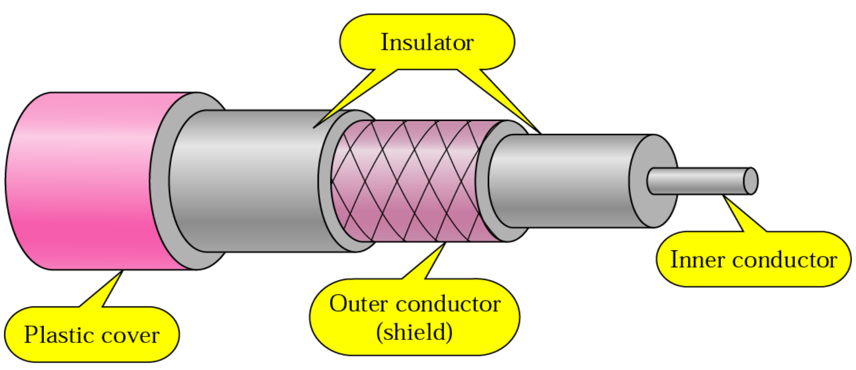



- Speed and throughput: 10-100 Mbps
- Cost per node: Moderately expensive
- Media and connector size: Medium to Large
- Maximum cable length: 100m (short)

- STP cable has a **metal foil** or **braided-mesh** covering that enhances each pair of insulated conductors.
- The metal casing prevents the penetration of electromagnetic noise.
- Materials and manufacturing requirements make STP more **expensive** than UTP but **less susceptible to noise**

# Coaxial cable

- Coaxial cable is a copper-cored cable surrounded by a heavy shielding and is used to connect computers in a network.
- The outer metallic wrapping serves both as a shield against noise and as the second conductor which completes the circuit.
- High bandwidth but lossy channel.
- Repeater is used to regenerate the weakened signals.



# Coaxial cable

- **Standards**

**Table 7.2** *Categories of coaxial cables*

| *Category* | *Impedance* | *Use* |
|---|---|---|
| RG-59 | 75 Ω | Cable TV |
| RG-58 | 50 Ω | Thin Ethernet |
| RG-11 | 50 Ω | Thick Ethernet |

RG - Radio Government rating

# Coaxial cable

## Coaxial Cable Connectors

- To connect coaxial cable to devices, it is necessary to use coaxial connectors.
- The most common type of connector is the Bayone-Neill-Concelman, or **BNC connectors.**

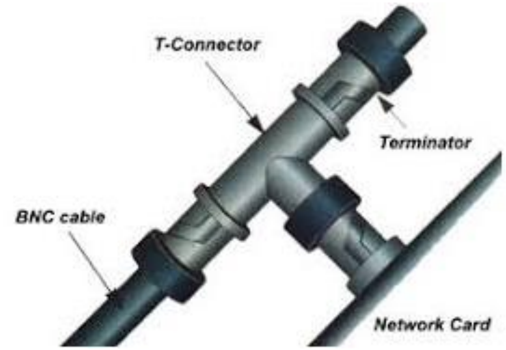
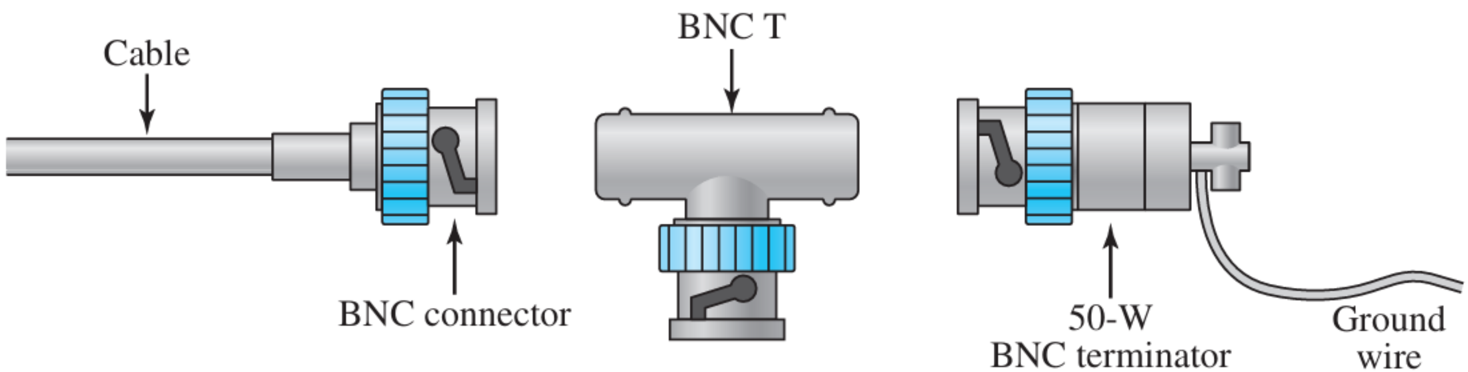



**Figure 7.8** *BNC connectors*



# Fiber Optics

- Made of glass or plastic, transmits signals in the form of light
- Light pulse can be used to signal a 1-bit and the absence of a pulse a 0 bit.

# Fiber Optics

- Optical transmission has 3 components:

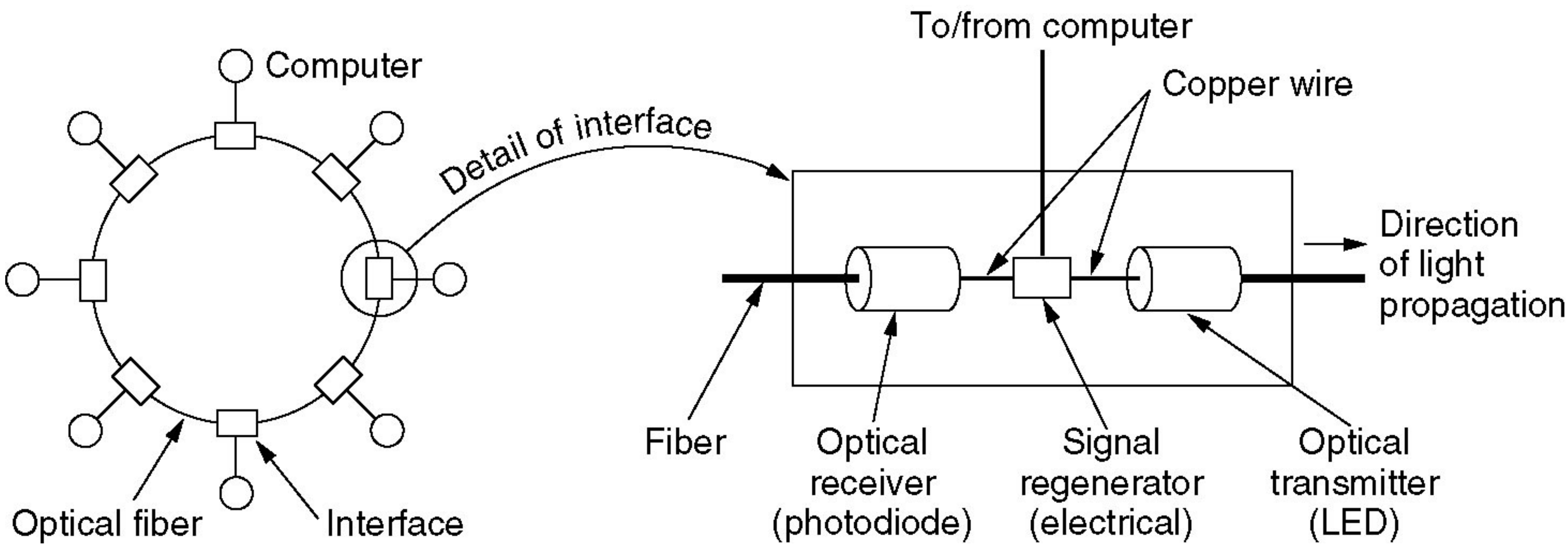
1. Transmission media (ultra thin glass fiber)
2. Light source (LED/laser diode) both emit light pulses when an electric current is applied.
3. Detector(Photodiode) : Generates an electric pulse when light falls on it.

- By attaching an LED/laser to one end of the optical fiber and a photodiode to the other, we can create a unidirectional data transmission system that

1. Accepts an electrical signal, converts it to light pulses and transmits these light pulses.
2. Reconverts the output to an electrical signal at the receiving end.

# Fiber Optic Networks

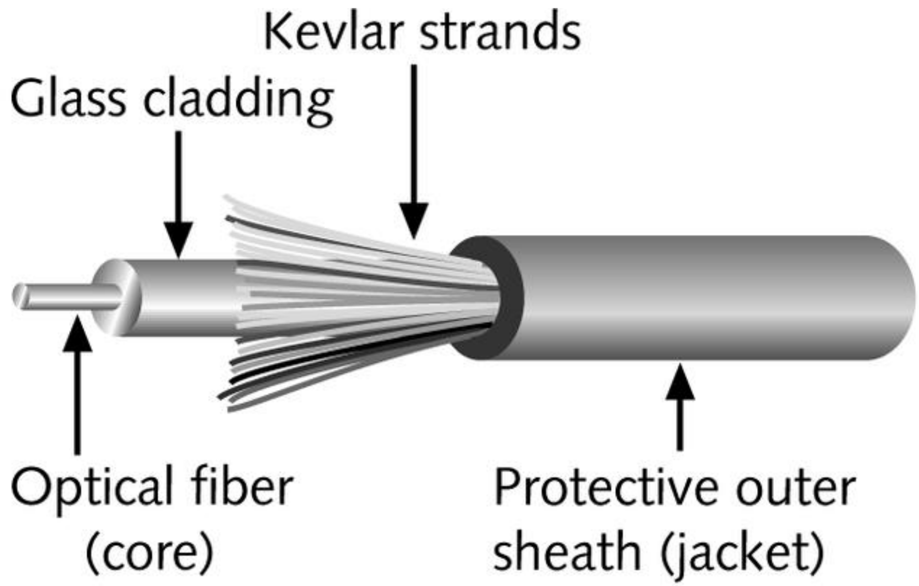
**A fiber optic ring (uni-drectional) with active repeaters.**

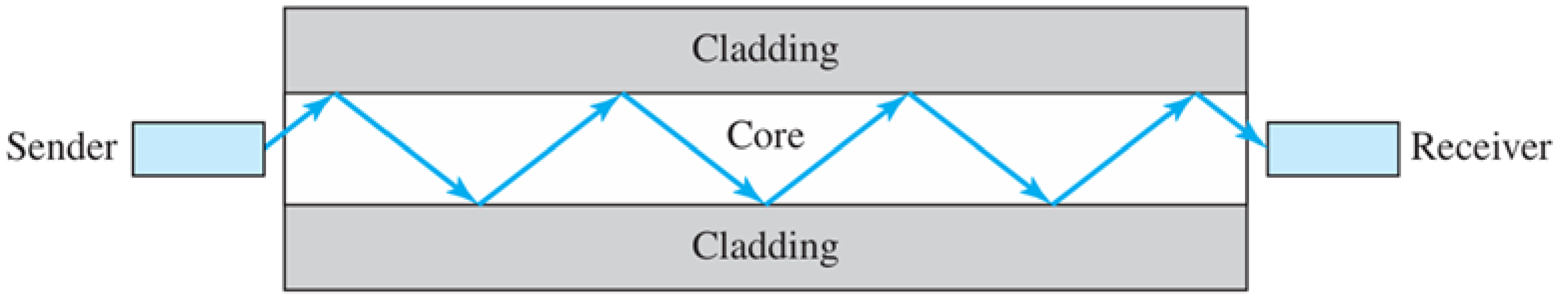


# Fiber Optics

## Structure

- Contains one or several glass fibers at its **core**
- Surrounding the fibers is a layer called **cladding**

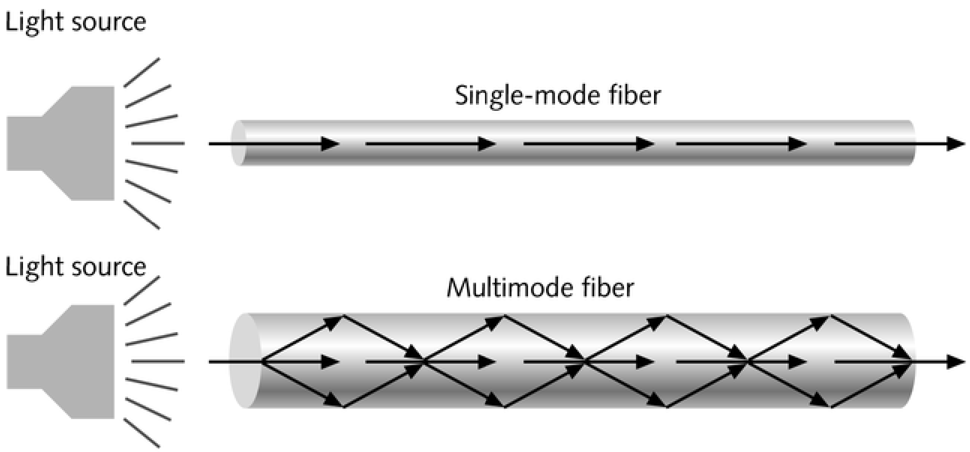


# Fiber Optics

## Types of fiber Optics

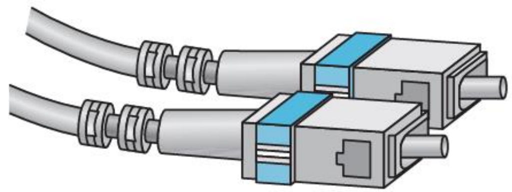
- **Single-mode fiber**
  - Carries light pulses along single path
  - Uses Laser Light Source
- **Multimode fiber**
  - Many pulses of light generated by LED travel at different angles



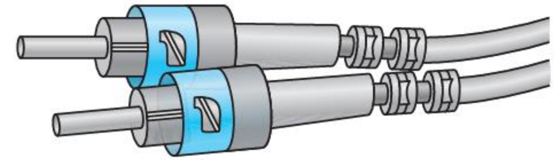
# Fiber Optics

## Connectors:

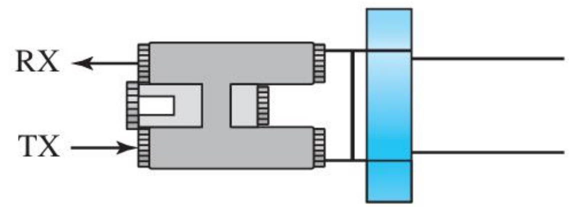
1. **Subscriber channel (SC) connector**
   - is used for cable TV
2. **Straight-tip (ST) connector**
   - is used for connecting cable to networking devices
   1. **MT-RJ**
   - is a connector that is the same size as RJ45



SC connector



ST connector

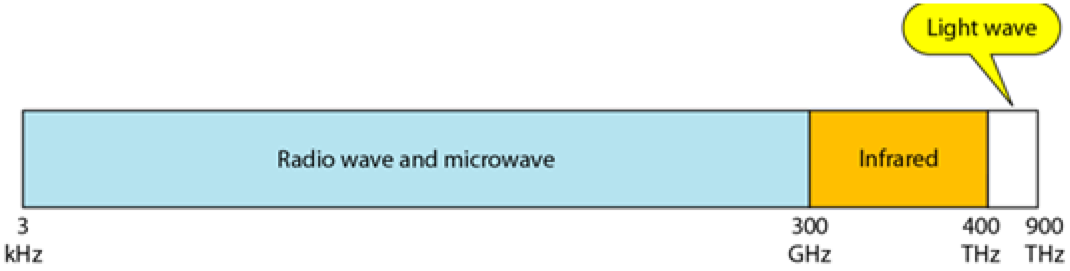



MT-RJ connector

# Unguided media (Wireless Communication)

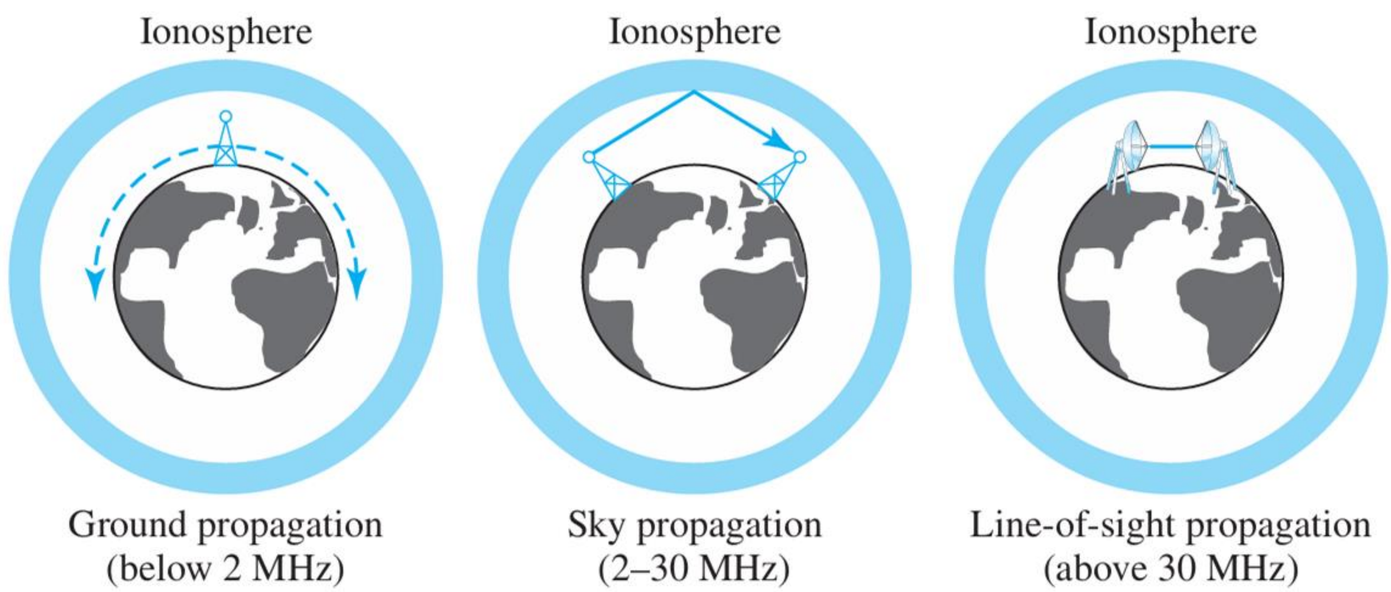
- **Drawback of Guided media:**

  Acceptable for short distance but expensive for long distance.
- Unguided media(or wireless communication) transport electromagnetic waves without using a physical conductor.
- Signals are broadcast through air (or, in a few cases, water), and thus are available to anyone who has a device capable of receiving them.
- How to generate electromagnetic waves - Inducing current in transmitting antenna.




3kHz-1 GHz - **Radio Waves**
1 GHz-300 GHz - **Microwaves**

# Unguided media (Wireless Communication)




**ground propagation :** , radio waves travel through the lowest portion of the atmosphere, hugging the earth. These low-frequency signals emanate in all directions from the transmitting antenna and follow the curvature of the planet. Distance depends on the amount of power in the signal: The greater the power, the greater the distance.

**sky propagation**, higher-frequency radio waves radiate upward into the ionosphere (the layer of atmosphere where particles exist as ions) where they are reflected back to earth. This type of transmission allows for greater distances with lower output power.

**line-of-sight propagation**, very high-frequency signals are transmitted in straight lines directly from antenna to antenna. Antennas must be directional, facing each other, and either tall enough or close enough together not to be affected by the curvature of the earth.

It’s tricky because radio transmissions cannot be completely focused.

# Unguided media (Wireless Communication)

- The section of the electromagnetic spectrum defined as radio waves and microwaves is divided into **eight ranges**, called **bands**.

**Table 7.4** *Bands*

| *Band* | *Range* | *Propagation* | *Application* |
|---|---|---|---|
| very low frequency (VLF) | 3–30 kHz | Ground | Long-range radio navigation |
| low frequency (LF) | 30–300 kHz | Ground | Radio beacons and navigational locators |
| middle frequency (MF) | 300 kHz–3 MHz | Sky | AM radio |
| high frequency (HF) | 3–30 MHz | Sky | Citizens band (CB), ship/aircraft |
| very high frequency (VHF) | 30–300 MHz | Sky and line-of-sight | VHF TV, FM radio |
| ultrahigh frequency (UHF) | 300 MHz–3 GHz | Line-of-sight | UHF TV, cellular phones, paging, satellite |
| superhigh frequency (SF) | 3–30 GHz | Line-of-sight | Satellite |
| extremely high frequency (EHF) | 30–300 GHz | Line-of-sight | Radar, satellite |

# Radio Waves

- Electromagnetic waves ranging in frequencies between **3 KHz and 1 GHz** are normally called radio waves.
- Radio waves are **omnidirectional**. When an antenna transmits radio waves, they are propagated in all directions. This means that the sending and receiving antennas do not have to be aligned. A sending antenna send waves that can be received by any receiving antenna.

**Omnidirectional Antenna for Radio Waves**

- Radio waves use omnidirectional antennas that send out signals in all directions. They follow the curvature of the earth.

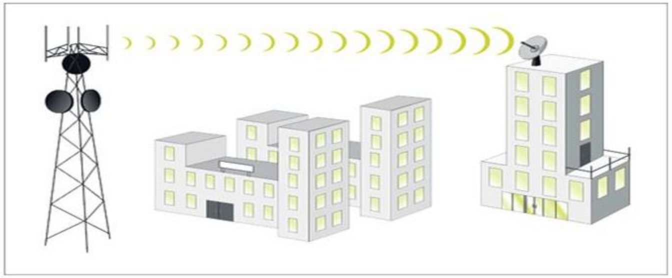
# Radio Waves

## Applications of Radio Waves

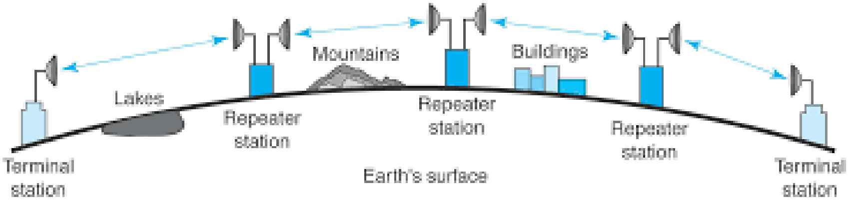
- The omnidirectional characteristics of radio waves make them useful for multicasting in which there is one sender but many receivers.
- AM(Amplitude modulation) and FM(Frequency modulation) radio, television, maritime radio, cordless phones, and paging are examples of multicasting.

# Microwaves

- Microwave transmissions do not follow the curvature of the earth.
- Requires line of sight transmission and reception equipment.
- Transmission/reception equipment if at a height of 100 m, distance between them can be max 82km.



To increase the distance---system of repeaters can be used




**Applications**

Cellular telephones, satellite communication and wireless LANs

# Infrared

- Infrared waves, with frequencies from 300 GHz to 400 THz, can be used for short-range communication.
- Infrared waves, having high frequencies, cannot penetrate walls.
- This advantageous characteristic prevents interference between one system and another, a short-range communication system in room cannot be affected by another system in the next room.
- When we use infrared remote control, we do not interfere with the use of the remote by our neighbours.
- However, this same characteristic makes infrared signals useless for long-range communication.
- In addition, we cannot use infrared waves outside a building because the sun's rays contain infrared waves that can interfere with the communication.

# Infrared

## Applications of Infrared Waves

- The infrared band, almost 400 THz, has an excellent potential for data transmission. Such a wide bandwidth can be used to transmit digital data with a very high data rate.